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GHG emissions and energy performance of offshore wind power

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ABSTRACT

This paper presents specific life cycle GHG emissions from wind power generation from six different 5 MW offshore wind turbine conceptual designs. In addition, the energy performance, expressed by the energy indicators Energy Payback Ratio (EPR) Energy Payback Time (EPT), is calculated for each of the concepts.

There are currently few LCA studies in existence which analyse offshore wind turbines with rated power as great as 5 MW. The results, therefore, give valuable additional environmental information concerning large offshore wind power. The resulting GHG emissions vary between 18 and 31.4 g CO_2-equivalents per kWh while the energy performance, assessed as EPR and EPT, varies between 7.5 and 12.9, and 1.6 and 2.7 years, respectively. The relatively large ranges in GHG emissions and energy performance are chiefly the result of the differing steel masses required for the analysed platforms. One major conclusion from this study is that specific platform/foundation steel masses are important for the overall GHG emissions relating to offshore wind power. Other parameters of importance when comparing the environmental performance of offshore wind concepts are the lifetime of the turbines, wind conditions, distance to shore, and installation and decommissioning activities.

Even though the GHG emissions from wind power vary to a relatively large degree, wind power can fully compete with other low GHG emission electricity technologies, such as nuclear, photovoltaic and hydro power.

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1. Introduction

All electricity generation technologies consume energy and emit greenhouse gases (GHGs) to a greater or lesser degree. When assessing the environmental performance of electricity generation it is important to take a Life Cycle Assessment (LCA) approach. This enables assessment of both the investment and the operating impacts relating to the generation process, and means that the entire life cycle of the investigated power plant, including upstream and downstream processes, should be taken into consideration. Upstream processes include, for example, mining and transport activities relating to the extraction of fuel, as well as extracting and processing activities relating to the materials used for building the power plant. Typical downstream processes include activities related to building and operating the grid, as well as the management of waste from the power generation processes. For most renewable electricity technologies and nuclear power, upstream and downstream GHG emissions account for over 90% of the cumulative GHG emissions. For conventional fossil fuel technology, however, the upstream GHG emissions also impact on the total picture, as they can represent up to 25% of the direct emissions from the power generation [1].

According to the IPCC Special Report on Renewable Energy Sources and Climate Change Mitigation [2], wind energy offers significant potential for the reduction of near-term (2020) and long-term (2050) greenhouse gas (GHG) emissions. This is achieved by generating electricity from larger, grid-connected wind farms, deployed either on- or offshore. At the end of 2009, the total installed wind power capacity of 160 GW, of which 2.1 GW comprised offshore capacity, was capable of meeting roughly 1.8% of worldwide electricity demand. This contribution could increase to about 20% by 2050 if ambitious efforts were made to reduce GHG emissions and to address the other limiting factors for large-scale wind energy development [2].

Wind turbines with a rated power of 5–6 MW are now being designed and installed, mostly for offshore operation [3]. There seem, however, to be few available studies concerning the environmental assessment of these ratings in relation to offshore turbines. Weinzettel et al. [4] have analysed the environmental performance of a floating 5 MW offshore wind turbine (Sway

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concept), and Tveten [5] has analysed wind power generation based on 5 MW offshore turbines in Scandinavia. Schleisner [6], Voorspools et al. [7], DONG Energy [8], Jungbluth et al. [9], Bauer et al. [10], Chataignere and Le Boult [11] and Vestas ([12] and [13]) have all assessed offshore wind power LCAs with turbine ratings from 0.5 MW to 3 MW. In the case of onshore wind power, however, there are several existing studies [14–21].

The aim of this paper is to present LCA GHG emissions and energy performance of six different offshore 5 MW wind power conceptual designs. The paper focuses on exploring the variations of the concepts rather than making a detailed ranking of the various different concepts. In addition, comparisons with relevant wind power LCA data are presented. The work has been carried out as a part of the research project Energy Trading & Environment 2020 [22].

The paper is organised as follows: Section 2 gives a short presentation of the Life Cycle Assessment methodology and the investigated energy indicators. Section 3 describes the offshore conceptual designs which have been investigated, while the resulting GHG emissions and energy performance are presented in Section 4. The results are compared with relevant literature data in Section 5, while Section 6 presents the conclusions.

2. Life Cycle Assessment (LCA) methodology

Life Cycle Assessment (LCA) represents a structured, comprehensive and internationally standardised (ISO 14044:2006 [23]) method for quantifying environmental and health impacts, resources consumed and resource depletion associated with any goods or services. In accordance with the International Reference Life Cycle Data System (ILCD) Handbook [24], Life Cycle Thinking and LCA create the scientific approaches behind modern environmental policies and business decision support relating to sustainable production and consumption.

Every electricity technology has an outage probability, whether it consists of a system of geographically dispersed wind farms, a hydro power station with a reservoir, or a fossil-fuelled power plant. The effect of adding new capacity can be quantified by the capacity credit. This is the capacity of conventional plants displaced by the new capacity, with an unchanged probability of failure to meet the reliability criteria of the system [25]. With high penetration levels of renewable energy, the capacity credit of different technologies such as wind energy, solar energy and bio energy could differ significantly. These differences have been ignored in this study. This simplification does not affect the comparison between the various different offshore wind turbine conceptual designs, but should be taken into account when comparing wind, solar and conventional energy technologies.

2.1. Analysed environmental indicators

This paper presents the environmental indicators GHG emissions and energy performance related to wind power generation. The GHG emissions have been calculated as Global Warming Potential (GWP), presented as g CO_2-equivalents. With regard to energy performance, two of the most common energy indictors for renewable electricity generation have been calculated: Energy Payback Ratio (EPR) and Energy Payback Time (EPT). A short description of these indicators is given below.

Energy Payback Ratio (*EPR*) expresses the amount of delivered energy during the power plant's lifetime, per energy unit invested in infrastructure and extraction/transport processes. It should be noted that the literature uses various different expressions for the EPR indicator. Examples of these are 'energy ratio', 'external energy ratio' and 'energy return on investment (EROI)', all of which refer to the same basic calculation as EPR [26]. In accordance with Hall [27], the EPR indicator refers to the amount of energy returned from one unit of energy invested in an energy-producing activity. A high EPR value means high energy efficiency. It should be mentioned that the energy being included in the fuel which represents the energy source (such as coal or gas) for thermal power plants is not included as invested energy in EPR calculations. This makes comparisons difficult between thermal and non-thermal electricity technologies due to the relatively high losses in the electricity conversion step for thermal power generation.

Energy Payback Time (*EPT*) expresses the amount of time in months or years, taken to "pay back" the energy invested in infrastructure and extraction/transport processes. A low EPT value means high energy efficiency. As in calculations for EPR, the energy being included in the fuel which represents the energy source is not included as invested energy in the calculation of EPT.

EPR represents a good energy indicator for assessing whether a wind turbine actually produces more energy than it consumes during its life cycle. EPT, on the other hand, measures the amount of electricity-producing months or years, which are required in order to pay back the energy invested in the wind power plant. It should be emphasised that EPR is dependent on the lifetime assumed for the power plant while EPT is independent of this parameter. The relationship between the parameters is expressed using the following equation:

$$\mathrm{EPR} = \mathrm{Lifetime}/\mathrm{EPT} \tag{1}$$

Raadal et al. [28] present a detailed investigation and discussion of different energy indicators for electricity generation.

3. The investigated offshore wind power concepts

Table 1 shows the analysed six offshore wind turbine concepts, comprising five floating and one bottom-fixed.

All concepts use the NREL 5 MW offshore reference wind turbine Rotor-Nacelle-Assembly (RNA), based on Jonkman et al. [33]. The hub height is 90 m and the rotor diameter is 126 m. The water depth is 200 m for the floating concepts and 50 m for the bottom-fixed concept. The wind farm (bottom-fixed or floating) is assumed to be located 200 km off the British Coast, at Doggerbank (independent of the real water depth), and consists of 100 wind turbines installed in a square layout (10*10 turbines). Fig. 1 illustrates the different concepts.

The Sway Tension-Leg Spar (TLS) is a single spar with excess buoyancy, one vertical tendon and a downwind turbine. The tower structure utilises external axial stiffening rods. More information can be found in Ref. [29].

The UMaine Semi-Submersible concept was developed in the DeepCwind project at the University of Maine. The concept consists

Table 1
Overview of the analysed concepts.

Concept	Name	General description	Reference
Floating	SWAY	Tension-Leg-Spar (TLS) similar to the SWAY concept	Borgen [29]
	UMaine Semi-S	UMaine Semi-Submersible	Robertson and Jonkman [30]
	UMaine Spar	UMaine Spar-Buoy (same as OC3-Hywind, at water depth of 200 m)	
	UMaine TLP	Tension-Leg-Platform with vertical tendons	
	MIT TLB	MIT Tension-Leg-Buoy (TLB),	Sclavounos et al. [31]
Bottom-fixed	OC4 Jacket	IEA OC4 Jacket	Vorpahl et al. [32]

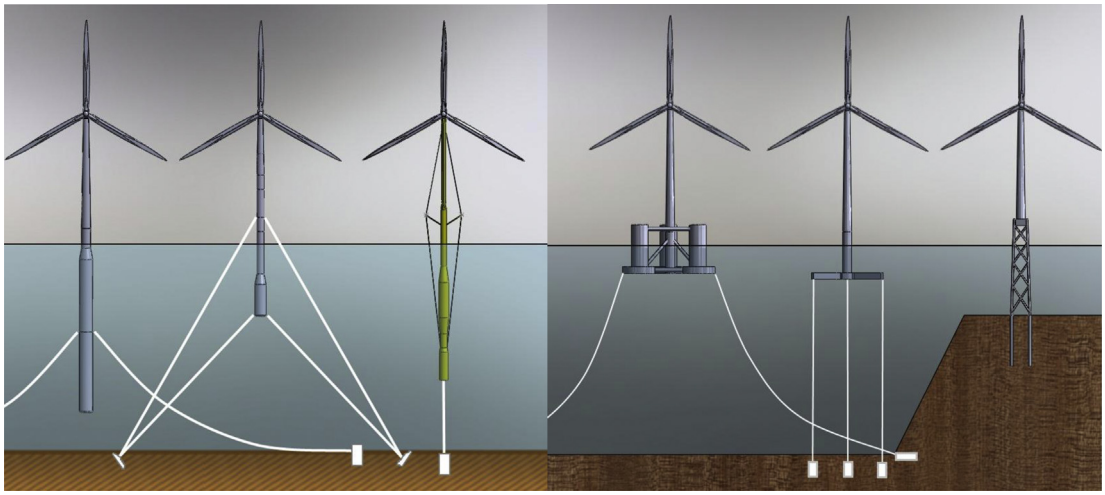

Fig. 1. Illustration of the concepts, from left to right; UMaine Spar, MIT TLB, SWAY, UMaine Semi-Sub, UMaine TLP and OC4 Jacket.

of a generic semisubmersible platform with the tower on a central column and three larger offset vertical columns connected by slender pontoons. The column stabilised platform has an overall draft of 20 m and utilises a catenary mooring system, water ballast and heave damper plates [30].

In the IEA OC3 project, the OC3-HYWIND was defined and made available to the research communities [34]. The UMaine spar buoy is essentially the same, but applied at a water depth of 200 m with a mooring system designed for that depth. The OC3-HYWIND was again largely based on the HYWIND concept and combined with an adapted version of the NREL 5 MW tower and turbine. The spar buoy relies on ballast stabilisation, deep draft and a catenary mooring system with crow-foot delta connections.

A small and relatively low mass TLP system is represented by the UMaine TLP consisting of a ballasted cylindrical platform with three horizontally extended legs, supporting the vertical tethering tendons, from its base. The concept relies on excess buoyancy to ensure taut tendons as in both the Sway TLS concept and MIT TLB.

The MIT TLB is a hybrid platform based on the same concept as Sway TLS but with a different solution for the mooring system. Two sets of inclined mooring lines are attached at two heights. One set of mooring lines has crow-foot deltas to control the yaw motions of the platform. All platform degrees-of-freedom are thereby stiffness-controlled, and all eigen periods are below the energetic part of the wave spectrum [31].

The OC4 Jacket is a space-frame installed at 50 m depth. The transition piece connecting the platform and the tower is made of concrete. The structure was designed for the IEA OC4 project [35].

The default wind farm is assumed to have a 20-year lifetime and a Capacity Factor (CF) of 46%. The CF expresses the actual annual electricity generation divided by the maximum possible annual electricity generation (operating at full power). It is expressed as a fraction, or a percentage. The functional unit for the analyses in this study is 1 kWh electricity generated and fed to the grid onshore. Thus, the GHG emissions and energy performance correspond to the generation of 1 kWh of wind power delivered to the grid onshore. The system boundaries include all the relevant life cycle stages, e.g. production of raw materials, transport, installation and decommissioning etc. Recycling credits from end of life treatment have not been included. This is in line with the Product Category Rules for electricity generation, in accordance with the International EPD System [36]. Grid losses through cables from offshore to onshore have not been included.

Table 2 lists the masses for the components above the substructure. The interface between the substructure and the tower is 10 m above the still water line. Table 3 lists the masses for the different substructures.

Low density concrete (2190 kg/m^3) has been used as grout material (included in the bottom-fixed foundation), while a density of 2380 kg/m^3 has been assumed for standard offshore concrete. The Ecoinvent processes “Poor concrete, at plant/CH U” and “Concrete, normal, at plant/CH U” have been used, respectively [37]. The applied steel density is 7850 kg/m^3 (the analyses do not adjust for bolts and flanges). Coatings are not included in infrastructure. For standard offshore steel and rebar steel, the Ecoinvent processes “Steel, converter, low-alloyed, at plant/RER U” and “Steel, low-

Table 2
Materials and masses for the adjusted NREL 5 MW offshore turbine [33] and corresponding Ecoinvent processes [37] used in the analyses.

Turbine NREL 5 MW (all values in tons per WT)	Rotor (hub and blades)	Nacelle	Tower structure	Sum	Ecoinvent processes used in the analyses [37]
Steel	60	197	234	490	Steel, converter, low-alloyed, at plant/RER U (rotor and nacelle), Steel, low-alloyed, at plant/RER U (tower)
Aluminium		8	5	13	Aluminium, primary, at plant/RER S
Electronics			4	4	Electronics for control units/RER U
Plastic			4	4	Polyethylene terephthalate, granulate, bottle grade, at plant/RER U
Copper		32	2	35	Copper, at regional storage/RER U
Oil			2	2	Lubricating oil, at plant/RER U
Glass reinforced plastic	50	2		53	Glass fibre reinforced plastic, polyamide, injection moulding, at plant/RER U
Sum	110	240	250	600	

Table 3
Materials and masses for the various different foundation/platform concepts [29–32].

Foundation/platforms, anchor and mooring cables (tons per WT)	Steel					HDPE	Concrete		
	Platform		Anchor	Mooring cables	Total steel	Mooring cables	Platform		Total concrete
	(Standard offshore)	(Reebar)	(Standard offshore)				(Standard)	(Grout)	
OC4-Jacket	539	67	153		759		59	66	126
SWAY	1050		188	20	1258				
Umaine Semi-S	3000	2	45	294	3341		205		205
Umaine Spar	1600	1	60	204	1865		143		143
Umaine TLP	525		354	60	939				
MIT TLB[a]	400	25	121		546	78			

[a] Increased tower mass (reebar steel) due to upper mooring line tethering system.

alloyed, at plant/RER U" have been used, respectively [37]. For the HDPE mooring cable for the MIT TLB, the Ecoinvent "Polyethylene, HDPE, granulate, at plant/RER U" process has been used [37]. All the floating concepts use water as ballast material, except the Umaine Spar which uses olivine.

The type, composition and weight of the offshore internal cables, typically three-phase 30–36 kV cables, are provided by Nexans [38] and Kaiser & Snyder [39]. The external cable (high voltage transmission cables) represents 145 kV, based on data from Statoil's external cable at Sheringham Shoal [40]. Based on the provided data, respective production processes for the cable material have been constructed by the use of representative Ecoinvent materials and corresponding processes.

One 500 MVA offshore substation is assumed to support the 100 5 MW turbines. The data have been provided by upscaling one 220 MVA offshore substation based on Tveten [5].

The calculation of the acquired fleet, time and fuel consumption relating to installation and decommissioning has been based on a combination of information from Sanden & Vold [41], Nielsen [42], Reitan [43] and Stuart [44]. For reasons of simplicity, the run of the installation of the floating concepts has been assumed to be equal and consists of towing, stabilising and installing activities. The run of the installation of the bottom-fixed platform includes transport and installation. For the same reasons, the operational time for installing the bottom-fixed and floating wind turbine concepts is assumed to be the same. However, the bottom-fixed operational window is assumed to be 65% which is 5% higher than for the floating concept, due to more complex operations. Maintenance is also assumed to be the same for all the concepts and is based on data from Sanden and Vold [41].

With regard to decommissioning and waste treatment of the wind turbines after end-of-life, the following assumptions have been drawn:

- Concrete: 15% to landfill, 85% to recycling
- Steel: 10% to landfill, 90% to recycling
- Glass reinforced plastic (in nacelle and rotor): 100% to recycling
- Aluminium (in nacelle, tower and transformer): 10% to landfill, 90% to recycling
- Copper (in nacelle, tower and transformer): 10% to landfill, 90% to recycling
- Cables (internal and external): 66% to recycling and 34% to energy recovery
- HDPE (mooring cable for MIT TLB): 100% to energy recovery

In accordance with [36], the environmental burdens relating to recycling shall not be included in the analysis since these burdens shall be allocated to the user of the recycled material. Therefore, transport and waste treatment burdens are only included for landfill and energy recovery activities. Data for relevant Ecoinvent processes [37] have been used for these processes. Other important data and assumptions with regard to the analyses are described in detail in Raadal and Vold [45].

When considering the realisation of the conceptual designs for the floating wind turbines, the study's perspective has been that of industrialisation and the large scale deployment of hundreds, if not thousands of wind turbines. When taking this perspective, installation procedures and equipment can be tailored to the various different conceptual designs. Relatively low costs have been assumed for maintenance compared to those we see today because of the large scale perspective chosen for the study. Later in the paper, a sensitivity analysis on emissions during installation is put in perspective in relation to the other emissions, showing that this rather crude approach is a useful first approximation. When considering installation of the first offshore floating wind farms, the differences in installation will be much more significant. Some conceptual designs, such as the semi-submersible, can be assembled in port and towed out. For the MIT TLB, the installation procedures are not yet fully developed and proven. For more detailed studies, and for evaluation of the first wind farms, differentiation on installation and maintenance would be required.

4. Resulting GHG emissions and energy performance for offshore wind

4.1. GHG emissions

The presentation of the GHG emissions results has been separated into the life cycle stages as shown in Table 4.

Fig. 2 shows the resulting GHG emissions (grams CO_2-equivalents/kWh) for the investigated offshore wind farm concepts in relation to the life cycle stages described in Table 4.

As seen from the figure, the total GHG emissions from the investigated offshore concepts vary between 18.0 (MIT TLB) and 31.4 (Umaine Semi-S) g CO_2-equivalents/kWh, a difference which represents a 75% increase in relation to the MIT TLB concept (representing the lowest GHG emissions).

Further, the figure clearly shows that the turbine and foundation/platform materials contribute most to the overall GHG emissions. The platform contribution varies between 6.3 (MIT TLB) and 19.7 (Umaine Semi-S) g CO_2-equivalents/kWh, corresponding to 35% and 63% of the total GHG emissions from each installation, respectively. The variations between the concepts are placed in this category, since the tower and RNA are identical. The different platform concepts are further analysed in order to investigate the most important parameters affecting these GHG emissions. This is presented in Fig. 3.

As seen from Fig. 3, the main contributor to the overall platform GHG emissions for all the concepts is steel production (separated into steel relating to the platform and anchor/cables, respectively). The steel production activity contributes from 50% to 89% of overall platform GHG emissions, depending on the concept. Generally,

Table 4
Short description of the life cycle stages included in the analyses.

Life cycle stage	Description
Installation (fuel)	Fuel consumption related to the transportation of all the equipment from shore to offshore site in order to install the wind farm.
Turbine materials	Production, processing and transport of all the infrastructure material related to the turbine production. Disposal of materials is also included (credits from material to recycling are not included).
Platform materials	Production, processing and transport of all the infrastructure material related the platform production, including production of internal and external (from offshore to shore) cables. Disposal of materials is also included (credits from material to recycling are not included).
Maintenance (fuel)	Fuel consumption related to the transportation from shore to offshore site due to maintenance.
Maintenance (infrastructure/reinvestment)	Production, processing and transport of all the material used for maintenance during the lifetime of the wind farm. The reinvestments needed are given annually and multiplied with the lifetime. Disposal of materials used for maintenance is included in this life stage.
Maintenance (others)	Production, processing and transport of support materials used for maintenance (oil, cotton etc.) and treatment of waste.
Decommissioning (fuel)	Fuel consumed for decommissioning the wind farm. For simplification the decommissioning is assumed equal to the installation (fuel) and operation (reversed). Disposal of materials is included in the life cycle stages Turbine and platform materials, respectively.

apart from the MIT TLB, the second largest platform contributor is the production of aluminium in connection with internal and external cables. Overall contribution from external cables is 1.95 g CO_2-equivalents/kWh.

As shown in Fig. 2, the GHG emissions relating to the turbine materials (assumed to be equal for all the concepts) are 4.7 g CO_2-equivalents/kWh, thus contributing to a range between 15% (Umaine Semi-S) and 26% (MIT TLB) of the total GHG emissions. The main parameters affecting the turbine GHG emissions are presented in Fig. 4.

The figure clearly shows that the main materials affecting the turbine GHG emissions are steel and glass fibre, contributing to 55% and 27%, respectively. In addition, it can be seen that the main components of the turbine (tower, rotor and nacelle) contribute about one third each to the total turbine GHG emissions.

From Fig. 2 it can also be seen that the installation and decommissioning life cycle activities contribute respectively 6.2 and 5.8 g CO_2-equivalents/kWh for the bottom fixed and floating concepts. This represents a range varying from 18% to 33% of total GHG emissions relating to each installation.

4.2. Sensitivity analyses GHG emissions

4.2.1. Sensitivity analyses capacity factor (CF) and lifetime (LT)

An increase in CF (better wind conditions or higher availability) or lifetime (LT) will reduce the GHG emissions per unit electricity (GHG emissions per kWh). The change relative to a reference is expressed as:

$$\text{GHG emissions per kWh}/\text{GHG emissions per kWh}_{\text{ref}} = \left(\text{CF}_{\text{ref}}/\text{CF}\right)*\left(\text{LT}_{\text{ref}}/\text{LT}\right) \quad (2)$$

Fig. 5 shows the relative GHG emissions per kWh as a function of the relative CF. The significance of CF can also be seen in traditional economic decision metrics such as Net Present Value (NPV). One of the pitfalls with NPV is that a large cost at the end of the project, such as decommissioning, can be obscured, especially if a high rate of return is used in the calculations. This can influence decisions exposing a company to considerable risk.

The capacity factor of different offshore wind farms is shown to vary between 29% and 53% in the literature [46]. These values represent relative CF's of 0.63 and 1.15, respectively (related to the default CF of 0.46), corresponding to a relative GHGE in the range

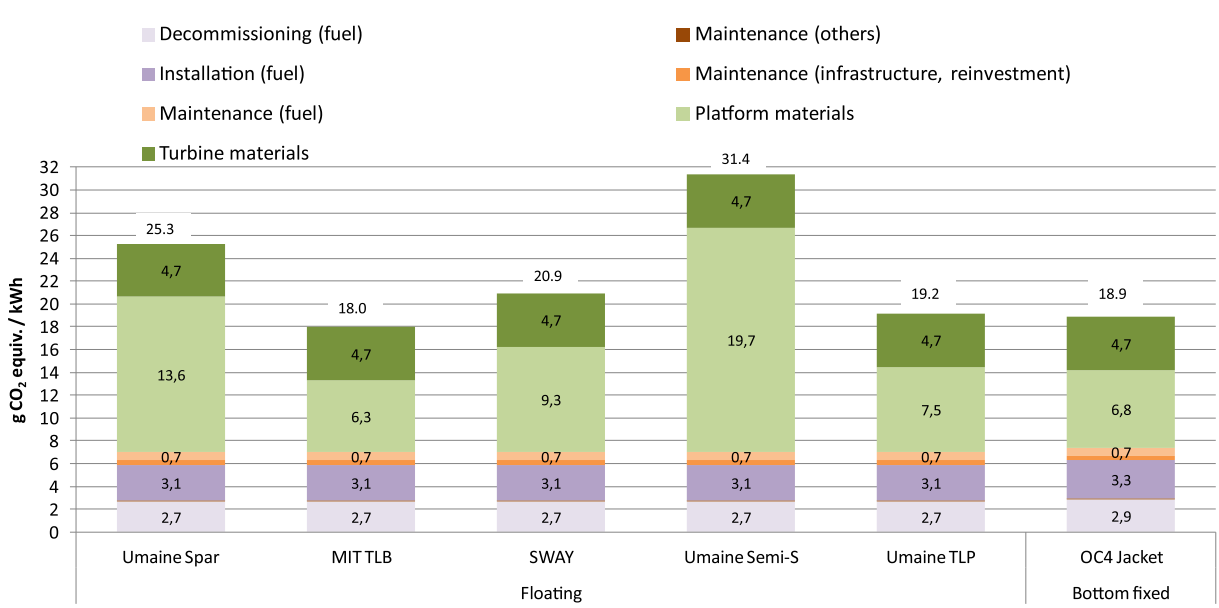

Fig. 2. GHG emissions (g CO_2-equivalents/kWh) for the investigated offshore wind farm concepts in relation to the different life cycle stages.

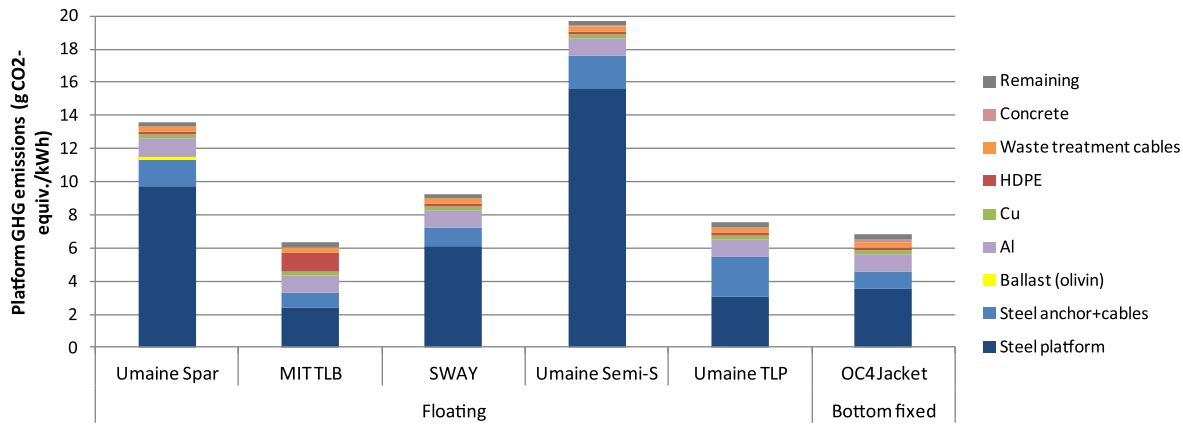

Fig. 3. Platform GHG emissions grouped into the main contributors.

0.87–1.58. Similarly, an increase of LT from 20 to 25 years (relative LT 1.25) will lead to a reduction in GHGE of 20% (relative GHGE emissions 0.8).

4.2.2. Sensitivity analysis with respect to steel masses

As described in Section 4.1, the materials relating to the foundation/platforms (including external/internal cables) are the main contributors to overall GHG emissions, representing a range from 35% to 63% of the total GHG emissions (dependent on the concept under consideration). Furthermore, from Fig. 3 it can be seen that the GHG emissions per kWh from the steel production of the foundation/platforms (see Table 3, excluding external/internal cables) represent between 36% and 79% of the total platform GHG emissions per kWh. Thus, the overall impact of the platform steel varies between 12% and 50% of the total offshore concepts' GHG emissions per kWh. This means that changing the platform steel amounts by 1% results in changes in GHG emissions of between 0.12% and 0.5% (depending on the concept under consideration). Fig. 6 shows the total GHG emissions dependent on the relative changes in platform steel masses. It should be emphasised that the two investigated cases OC4 Jacket and Umaine TLP cannot be clearly distinguished in the figure because they coincide.

As seen in Fig. 6, the Umaine Semi-S and the MIT TLP concepts are those which are, respectively, to the greatest and smallest degree, affected by a relative change in the platform steel masses. This is to be expected as these concepts represent the largest and smallest steel masses. The figure also shows that a significant change in the platform steel masses may alter the ranking of the conceptual design with regard to GHG emissions. This can be exemplified by a 53% reduction of platform steel masses in the Umaine Spar concept, which would result in GHG emissions per kWh equal to the Sway default concept.

This paper does not intend to rank the platforms according to GHG emissions per kWh, since all the conceptual designs with open data can be considered as 'first-generation designs'. New versions of floaters are being designed with significantly reduced steel masses as the industry gains experience, and one can similarly expect significant cost reductions for offshore floating wind turbines as those already seen for onshore wind turbines. The study nevertheless demonstrates the importance of overall steel masses on GHG emissions and energy performance.

4.2.3. Sensitivity analysis with respect to fuel consumption during installation and decommissioning

The installation and decommissioning life cycle stages represent the second largest contributor to the overall GHG emissions, ranging from 18% to 33% for the various different concepts. In order to investigate the robustness of the main assumptions in relation to these life cycle stages, a sensitivity analysis has been performed by increasing/decreasing the fuel consumption by 25%. The results are shown in Fig. 7.

As seen from the figure, a 25% increase/decrease in fuel consumption in relation to installation and decommissioning of the wind farms has comparatively little impact on the overall GHG

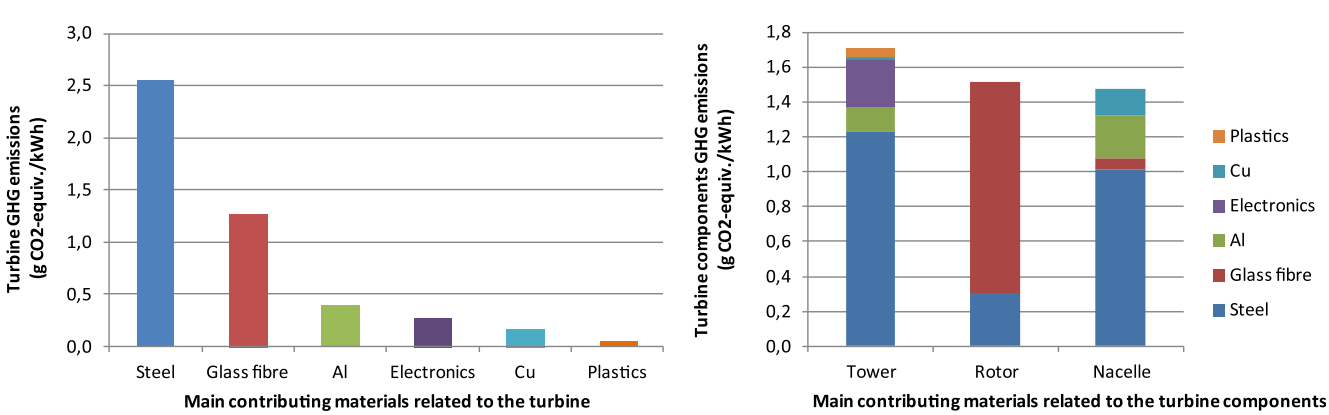

Fig. 4. Turbine GHG emissions grouped into the main contributors.

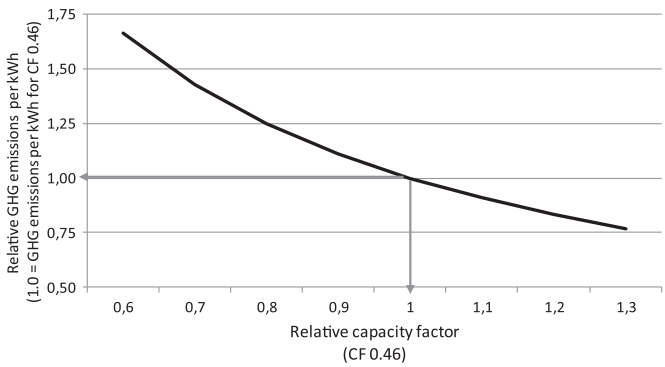

Fig. 5. Sensitivity analysis showing variations in relative GHG emissions per kWh as function of relative capacity factor (CF).

emissions. The increase/decrease in the overall GHG emissions ranges between 5% and 8% dependent on the analysed concept.

4.3. Energy performance

The energy indicators EPR (Energy Payback Ratio) and EPT (Energy Payback Time), as defined in Section 2.1, are presented in Fig. 8. The greater the EPR indicator value, the better the energy performance, while the opposite applies in the case of EPT (the smaller the value the better the performance).

The figure shows that the investigated concepts achieve EPR and EPT values of between 7.5 and 12.9, and 1.6 and 2.7 years, respectively. The MIT TLB and OC4 Jacket give the best performance, being in line with the results for GHG emissions. This is to be expected since the use of conventional energy within an analysed system generally represents the main contributor to GHG emissions.

5. Comparison of GHG emissions and energy performance with relevant data

The GHG emissions and energy performance results for the investigated offshore concepts have been compared with relevant LCA literature data in order to put the investigated wind power cases and concepts in perspective.

5.1. GHG emissions

The results for GHG emissions (g CO2-equiv/kWh) have been compared with relevant LCA data [4–11,30,36–42] and categorised according to turbine size and capacity factor.

The literature data have been adjusted as follows, in order to ensure comparability: data representing lifetimes other than 20 years ([9,14,15] (2 cases) and [16]) have been adjusted for 20 years, and data including benefit from recycling of materials [18,19] have been adjusted to exclude these benefits. Some cases [12,13] have been excluded from the comparisons because the benefits from recycling included in the analyses have been too challenging to adjust. It should be emphasised, however, that lack of transparency in the underlying assumptions in the literature can mean that some unmodified data may still be included in the comparisons. The results are shown in Fig. 9, presenting onshore and offshore data as blue and red bullets (in web version), respectively. The offshore concepts investigated in this study are highlighted by circles. It should be noted that the values of the two investigated concepts Umaine TLP and OC4 Jacket (representing 19.2 and 18.9 g CO_2-equiv./kWh, respectively) coincide in the figure and are therefore hard to distinguish from each other.

Fig. 9 shows that there is a tendency for a decrease in GHG emissions where there are, respectively, increased turbine size and capacity factors. This is in accordance with results from other studies [1,15,20,46,47]. Despite this general tendency, Fig. 9 shows that the variations in the data are still relatively large. Such variations can be explained by differing presuppositions concerning the wind farm. These can include lifetime; end-of-life treatment of wind turbine components; the energy invested to build and install the wind turbines; reinvestment rates and maintenance; whether current or future conditions have been analysed as well as the type of analysis[2] (Process analysis or Input–output analysis).

As described above, the data in Fig. 9 have been adjusted with regard to the two first mentioned parameters (lifetime and end-of-life treatment) in order to make them comparable. As regards the impact of the type of analysis on the results, GHG emissions reported for wind power appear to increase when changing from Process analysis to Input–output analysis [46]. This corresponds with the results from a multivariate regression analysis, examining the influence of methodology, scope and technological maturity [15]. They show that the results of the energy intensity (and GHG emissions) increase during a change from Process to Input–output analysis. The literature data presented in this study include both analysis types, and the variations may be affected by this. Another source of scatter in the data could be that wind farms are optimised for minimum energy costs and not minimum GHG emissions.

In addition, Fig. 9 shows that the offshore concepts investigated in this study (highlighted by circles) generally result in higher GHG emissions when compared with other offshore wind power studies. This is further investigated using a more detailed comparison of these concepts and relevant offshore data, as shown in Fig. 10 (the investigated cases are presented as red bullets, in web version).

As seen in the figure, there are only two comparable offshore cases with regard to turbine size (Weinzettel et al. [4] and Tveten [5], highlighted by black circles) as the others represent turbines smaller than 2.5 MW. This shows that the results of the six conceptual offshore designs investigated in this study give valuable additional environmental information regarding large offshore wind power. The two above mentioned cases represent a Sway concept [4] and a “representative offshore concept in Scandinavia” [5] with corresponding GHG emissions of 12.2 g and 20.6 CO_2-equivalents/kWh, respectively ([5] adjusted for 20 years lifetime).

In order to compare the results for equal capacity factors, the two cases have been adjusted for the default capacity factor used in this study (46%). The respective modified GHG emissions, presented in Fig. 10 (right part) by the “stars” depicted in the graph, become 14.1 [4] and 16.8 [5] g CO_2-equivalents/kWh. This shows that both cases result in lower GHG emissions when compared with the investigated conceptual designs. The distances to shore in the comparable cases are, however, 75% [4] and 85% [5] shorter when compared with the distance (200 km) in this study. In order to assess the importance of distance to shore, the Sway conceptual design in this study has been modified by assuming 50 km distance to shore (a 75% reduction). The resulting GHG emissions of 18 g CO_2-equivalents/kWh represent a reduction of 14%, thus showing that distance to shore also affects the results and should be taken into consideration when comparing offshore wind concepts.

[2] A process analysis assesses the value chain of the system in a holistic manner, focussing on all significant processes based on mass and energy relations (bottom-up approach). An input–output analysis (IOA) assesses the environmental aspects of products and services by dividing a product into its economic components (machinery, chemistry, services, etc.) as the basis for performance calculation (top-down approach).

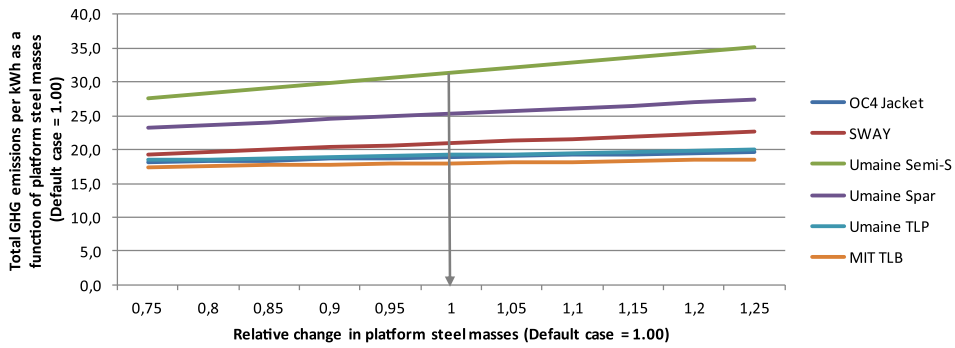

Fig. 6. Platform steel masses' impact on the offshore conceptual designs' overall GHG emissions.

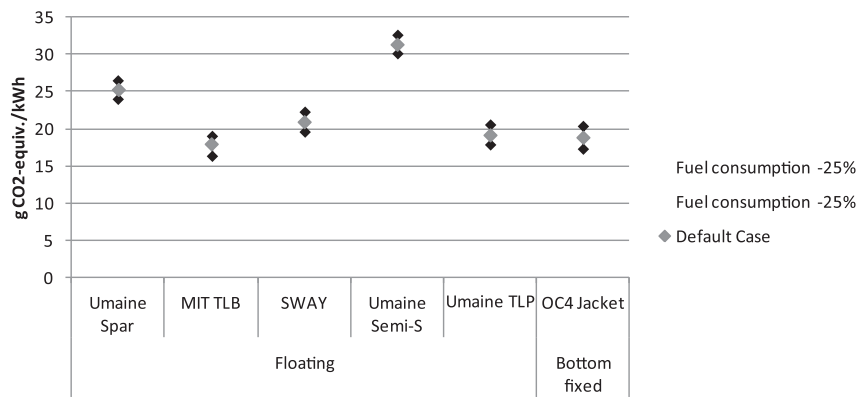

Fig. 7. Sensitivity analyses regarding fuel consumption for installation and decommissioning for the different offshore concepts.

The difference in the resulting GHG-emissions between the Sway case investigated in this study and the case described in Ref. [4] has been reduced from 33% to 22% by modifying equal conditions with regard to capacity factor and distance to shore. The difference in the remaining GHG emissions (22%) can partly be explained by the steel masses being 19% less in the Weinzettel et al. [4] case compared with the Sway concept analysed in this study. A more detailed analysis of this 19% difference in steel masses shows that the case investigated in Ref. [4] has not included steel masses relating to anchor and anchoring cables. In the Sway concept analysed in this study, these steel masses represent 210 tons, which accounts for 63% of the steel masses difference between the two Sway cases. Other differences between these cases which may be of importance for the results are transport to offshore (towing boats being the only vessels included in Ref. [4]), infrastructure maintenance (5% extra infrastructure material [4] when compared with 2%) and solid ballast material (included in Ref. [4]) when compared with water.

The comparisons and modification of the offshore concepts show that wind conditions and distance to shore are important factors to be taken into consideration when comparisons of offshore wind concepts are carried out. It confirms, furthermore, that the required steel infrastructure masses have great importance for the resulting GHG emissions.

It should be noted that Tveten [5] has used an Extended Input–Output LCA for the analysis. The use of these broader system boundaries would imply that increased GHG emissions could be expected, when compared with the Process LCA method used in

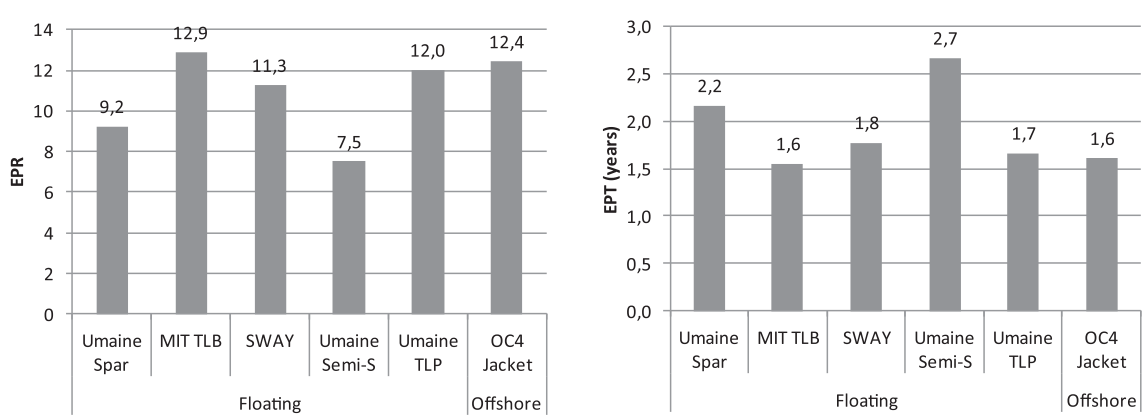

Fig. 8. Energy Payback Ratio (EPR) and Energy Payback Time (EPT) for the different offshore concepts.

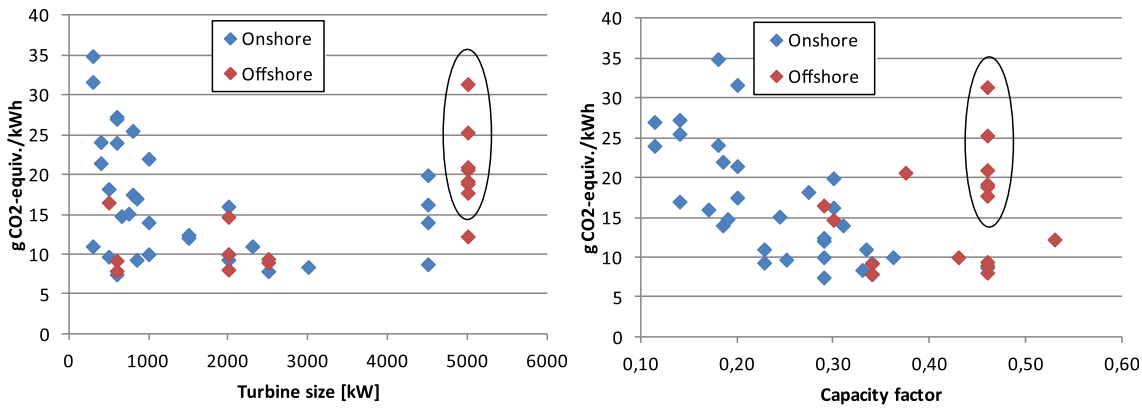

Fig. 9. Comparisons of GHG emissions of the investigated offshore cases with relevant LCA literature, categorised in accordance with turbine size (left) and capacity factor (right).

this study. This is, however, not the case; the principal reason being, in all probability, that, lacking specific data [48], Tveten [5] has chosen to simplify the platform data. Instead of using specific material and masses relating to the platform, a total mass of 1000 tons, comprising "foundation/ballast and mooring", has been applied. In addition, the data relating to this category have been based only on cost data for "proxy sector from the background system". Compared with the conceptual designs analysed in this study, the relating specific platform steel masses (including anchor and mooring cables) represent a range of between about 550 tons and 3350 tons (see Table 3). The corresponding GHG emissions represent a range of between 3.3 and 17.6 g CO_2-equiv./kWh, which account for 18%–56% of the total wind turbine GHG emissions. Thus it can be seen that specific platform steel masses have significance for the resulting GHG emissions.

Within the EU-supported NEEDS (New Energy Externalities Developments for Sustainability) project [8], projected future offshore wind turbine ratings have been considered. These are between 8 MW and 32 MW, and from 2005 until 2050. The scenarios represent various differing assumptions with regard to the development of future technology. Examples include improved steel production, increased share of renewables in the electricity mixes etc. for water depths varying between 10 and 30 m (2005), 20–60 m (2025) and >100 m (2050). The projected future technology models have been based on up-scaled data from the reference technology (Horns Rev Offshore Wind farm, 2 MW) and result

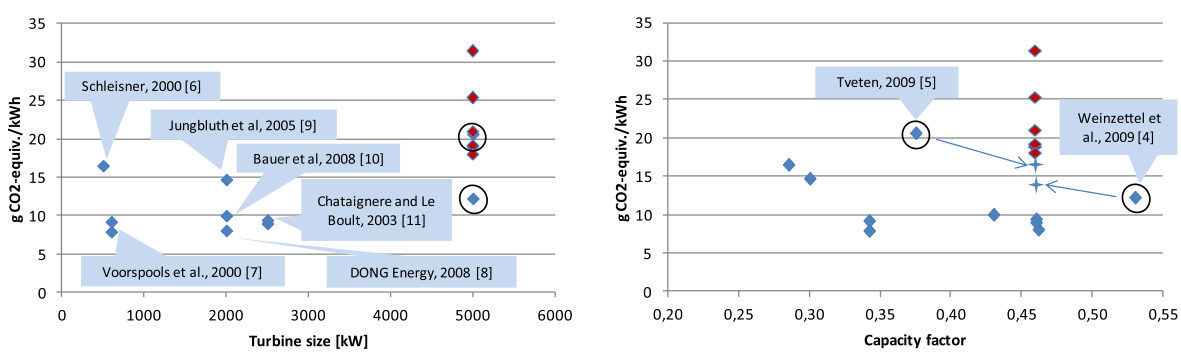

Fig. 10. Comparisons of the investigated offshore concepts with relevant LCA literature, categorised in accordance with turbine size (left) and capacity factor (right).

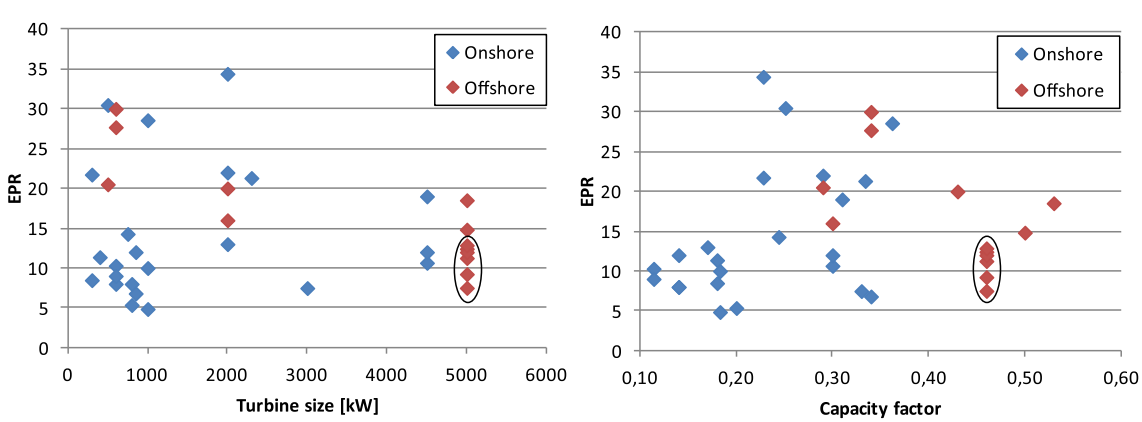

Fig. 11. Comparison of EPR (Energy Payback Ratio) for the investigated offshore cases with relevant literature, categorised in accordance with turbine size (left) and capacity factor (right).

in GHG emissions varying between 2.9 and 7.9 CO_2 -equivalents//kWh. The principal conclusions from this study are that the material used in the manufacture of wind turbine components is crucial for the overall environmental impact of a wind turbine.

5.2. Energy Payback Ratio (EPR)

In order to put the concepts in this study into perspective, the energy performance results or Energy Payback Ratio (EPR) have been compared with relevant LCA literature, just as with the GHG emissions.

The EPR results have been compared with relevant LCA data [4,6,7,9,10,30,37–41,45] and categorised in accordance with turbine size and capacity factor. The data have been standardised according to a 20 year lifetime and no credits from recycling have been included. The results are shown in Fig. 11, the offshore conceptual designs investigated in this study being highlighted by black circles. As emphasised for Figs. 9 and 10, the values of the two investigated concepts Umaine TLP and OC4 Jacket (representing 19.2 and 18.9 g CO_2-equiv./kWh, respectively) coincide in the figure and these concepts are therefore hard to distinguish from each other.

Lenzen and Munksgaard [15] confirm that small wind turbines of 1 kW require about three times more life-cycle energy per unit power than large wind turbines of 1 MW. Thus, it might have been expected that the EPR indicator would generally increase in accordance with increased turbine size (just as GHG emissions decrease with turbine size). It can be seen, however, from Fig. 11, that there exists a lack of correlation, both between EPR and turbine size and between EPR and capacity factor. This is in line with the conclusions drawn by Davidsson et al. [49], stating that there is an inconsistent use of terminology and concepts regarding wind power energy performance. The need for discussion on the way in which environmental impact, energy performance, and natural resource use of renewable energy recourses should be assessed is addressed by the authors [49]. The existence of this inconsistency in energy performance assessment is supported by Modahl et al. [50]. They conclude that assessing the energy indicator Cumulative Energy Demand (CED) for a product or service is more dependent on the LCA practitioner's knowledge and competence about the CED assessment method, than on other methods for assessing environmental indicators, e.g. carbon footprint. It is important to check whether resources are accounted for as material resources/feedstock or as energy resources. This is because the pre-defined CED methods which are available in LCA software tools often differ with regard to the challenges connected to feedstock issues. If they are not, then the energy resources may be handled differently, leading in turn to diverging results (Modahl et al. op.cit).

Davidsson et al. [49] point out another complicating issue with regard to energy performance assessment. Some studies include a conversion factor (reflecting the efficiency of the conventional electricity generation) relating to the produced electricity. This is in order to convert the produced wind power to the primary energy which would otherwise have been produced in a conventional power plant. There is no consensus, however, on how conversion to primary energy should be handled, or even if and why it should be done. It must be emphasised that the literature data in this study have been modified in order to exclude potential conversion factors, but lack of transparency may have resulted in unmodified data. The data have been modified by multiplying them with the given efficiencies related to the assumed conventional power plants. Davidsson et al. [49] summarise that the use of different methods, lack of transparency, and variations in the presentation of results, e.g. including and not including a conversion factor, make it difficult to compare studies with each other and with other renewable sources. The "disturbed picture" of energy performance shown in Fig. 11 confirms the above mentioned inconsistency in energy performance assessment.

Fig. 11 also shows that, the investigated offshore concepts generally show lower values than the offshore literature data, which means lower energy performance. This is in line with the results for the GHG emissions, as shown in Section 5.1.

6. Conclusions

There are currently few LCA studies in existence which analyse offshore wind turbines with rated power as great as 5 MW. The results, therefore, from the investigation into the six conceptual offshore designs used in this study, give valuable additional environmental information concerning large offshore wind power. The resulting GHG emissions vary between 18 and 31.4 g CO_2-equivalents/kWh while the energy performance, assessed as EPR and EPT, varies between 7.5 and 12.9, and 1.6 and 2.7 years, respectively. The relatively large ranges in GHG emissions and energy performance are chiefly the result of the differing steel masses required for the analysed foundation/platforms.

In comparison with the relevant existing offshore studies, the six conceptual designs result in larger GHG emissions and lower energy performance. The principal reason for this is that more complete and accurate data for steel masses relating to the platform, anchor and mooring cables have been included in the analysis. Thus, one major conclusion from this study is that specific platform/foundation steel masses are important for the overall GHG emissions relating to offshore wind power. Other parameters of importance when comparing the environmental performance of offshore wind concepts are the lifetime of the turbines, wind conditions, distance to shore, and installation and decommissioning activities.

The offshore wind power conceptual designs which have been investigated here, give larger GHG emissions and a lower energy performance when compared with the corresponding average values of the onshore wind power data. This can indicate that the extra energy invested in offshore plants, as a result of the need for additional platform infrastructure, may not be beneficial.

This study confirms the results from other research [15] which shows a tendency for GHG emissions to decrease in relation to an increase in turbine size and capacity factor. It should be noted, however, that despite this general tendency, relatively large variations within the data still remain. These variations can mainly be explained by the following: varying assumptions/conditions relating to the energy invested to build and install the wind turbines; the type of analysis (Process LCA or Input/Output); whether the turbine is located on or offshore, and varying reinvestment rates and maintenance activities.

In the case of energy performance of wind power, the study confirms the inconsistencies with regard to energy performance assessment methods which have been documented by Davidsson et al. [49] and Modahl et al. [50]. There is a need for an increased effort in work on standardising energy performance calculations methods.

Even though the GHG emissions from wind power vary to a relatively large degree they are small in relation to those from fossil electricity generation technologies. Wind power can fully compete with other low GHG emission electricity technologies, such as nuclear and photovoltaic power, and hydro power. GHG emissions from wind farms are small in relation to those from fossil electricity generation technologies.

Lastly, it should be emphasised that GHG emissions and energy performance represent only two environmental indicators. With regard to decision making and guiding policy, several other environmental indicators need to be taken into consideration. These

include land use, visual aspects, biodiversity and noise. This is particularly relevant when comparing onshore and offshore turbines.

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